Team Task Tracker with Role-Based Permissions and Activity Timeline

Objective

Build a mini Team Task Tracker web application that allows users to create teams and invite members, assign tasks within teams, and track updates (status changes, comments) in a chronological timeline.

Functional Requirements

Authentication & Users

- Implement user registration and login (JWT-based or session-based).
- Each user has: name, unique email, password (hashed), and a role (Admin or Member).
- Only Admins can create new teams or invite members.

Teams

- A team has: name, description, created_by (User ID), and list of members (User references).
- Admins can invite members via email (simulate invitation).

Tasks

- A task has: title, description, assigned_to (User ID), status (Pending / In Progress / Done), created_at, updated_at, team_id.
- Only team members can see and modify tasks within their team.

Activity Timeline

- Every time a task is created, updated, or commented on, an activity entry should be added to a chronological timeline.
- Example: ‘Saumyakanta changed status of “Design API endpoints” to In Progress.’
- Timeline should be visible on the team’s dashboard, ordered by most recent.

Tech Requirements

Backend (Django + MongoDB)

- Use Django (preferably Django REST Framework or GraphQL).
- Store data in MongoDB (using djongo or mongoengine).
- Implement proper models for User, Team, Task, and Activity.
- Use role-based permissions (custom middleware or decorators).
- Expose RESTful APIs for authentication, teams, tasks, and activity timeline.

Frontend (Next.js)

- Build UI using Next.js.
- Include pages: Login/Register, Dashboard, Team View, and Task Management.
- Show live updates on the timeline via polling or optimistic UI.

Evaluation Criteria

Backend Design, MongoDB Usage, Auth & Permissions, Frontend Integration, UI/UX, Problem-Solving Approach, and Code Quality.

Bonus Points

- Add task comments (with timestamps and user refs).
- Add filters for tasks by status.
- Use JWT refresh tokens and handle expiry on frontend.
- Use TailwindCSS or React Bootstrap for UI.